

UNIT 2

1 Mark Questions

1. What does IPR stand for in Cyber Laws?

→ *Intellectual Property Rights.*

2. Expand COBIT in the context of security governance.

→ *Control Objectives for Information and Related Technology.*

3. Define Security Assurance in one sentence.

→ *It is the confidence that security measures protect systems and data as intended.*

4. Mention any one international standard for information security.

→ *ISO/IEC 27001.*

5. What is the primary objective of a Security Audit?

→ *To evaluate and verify the effectiveness of security controls.*

2 Mark Questions

1. Define Security Assurance in the context of cyber laws.

- *Security Assurance* is the process of producing evidence (through testing, evaluation and review) that security controls are correctly implemented and effective in protecting systems and data to meet legal, regulatory and organizational requirements.
 - *Example:* Pen-tests and compliance checks that demonstrate adherence to a data-protection law.
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2. What is Intellectual Property Rights (IPR)? Give one example.

- *Intellectual Property Rights (IPR)* are legal rights that protect creations of the mind (e.g., inventions, literary and artistic works, designs, trademarks), giving creators exclusive control over use and distribution for a limited time.
 - *Example:* Copyright on software source code.
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3. Name any two international standards used in cybersecurity.

- *ISO/IEC 27001* — standard for Information Security Management Systems (ISMS) requirements.
 - *ISO/IEC 27002* — code of practice that provides guidelines for implementing information security controls.
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4. What is the role of a security audit?

- The role of a *security audit* is to systematically assess security policies, controls and procedures to verify compliance, identify weaknesses or gaps, and provide recommendations to reduce risk and improve security posture.
 - *Example:* An internal audit that verifies controls required for ISO/IEC 27001 certification.
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5. Expand COBIT and briefly state its purpose.

- **COBIT — Control Objectives for Information and Related Technology.**
 - *Purpose:* A governance and management framework that helps organizations align IT processes with business goals, define control objectives, assign responsibilities, and measure IT performance and compliance.
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4 Mark Questions

1. Explain the difference between Security Assurance and Security Compliance with suitable examples.

- **Security Assurance** is the process of building and providing objective evidence that security controls are correctly designed, implemented, and effective in reducing risks — it's about *confidence* in security. Methods include threat modelling, penetration testing, formal verification, and continuous monitoring.
- **Security Compliance** means meeting specific legal, regulatory or standards-based requirements (e.g., GDPR, ISO/IEC 27001, PCI-DSS); it is *conformance* to externally defined rules or checklists. Compliance shows that minimum controls are in place, but passing compliance checks does not guarantee real security.

- *Example:* A company may be **compliant** with ISO/IEC 27001 (has documented controls and audit reports) but still improve **assurance** by running red-team exercises and continuous monitoring to demonstrate real-world effectiveness.
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2. Describe the importance of cyber laws in preventing digital crimes.

- Cyber laws create the legal definitions, penalties, and procedures needed to deter, investigate, and punish digital offences (e.g., hacking, identity theft, data breaches), thereby raising the cost of cybercrime.
 - They enable lawful evidence collection, cross-border cooperation (MLATs and mutual legal assistance), takedown of illicit content, and remedies for victims — all of which make prosecution and remediation feasible.
 - By setting obligations for organizations (data protection, breach notification, liability), cyber laws also push businesses to adopt better security practices and accountability, reducing the overall risk of digital crimes.
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3. What are the major components of an effective Security Audit process?

- **Scope & Objectives:** Define assets, systems, standards, and audit goals (what's covered and why).
 - **Planning & Risk Assessment:** Identify critical systems, threats, and controls to prioritize audit activities.
 - **Evidence Collection & Testing:** Inspect configurations, logs, policies, and run tests (vulnerability scans, control checks, interviews).
 - **Evaluation & Findings:** Compare evidence against standards/policies, identify gaps and risk severity.
 - **Reporting & Recommendations:** Produce a clear report with prioritized remediation actions, timelines, and owners.
 - **Remediation Follow-up & Closure:** Verify fixes, re-test, and document closure; maintain audit trail for compliance and continuous improvement.
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4. Write a short note on SSE-CMM and its application in cybersecurity.

- **SSE-CMM (Systems Security Engineering Capability Maturity Model)** is a process reference/maturity model for security engineering that defines best practices and capability levels for organizations performing security engineering. Its goal is to make security engineering a measurable, repeatable discipline by describing process areas and maturity levels.
 - The model helps organizations assess and improve how they plan, design, implement, verify, and manage security across the system lifecycle; it is used to choose qualified providers, guide process improvement, and augment assurance activities.
 - **Application:** SSE-CMM is applied to evaluate security engineering capability (e.g., secure requirements, risk analysis, verification), guide investments in training/tools, and raise process maturity so that security becomes systematic rather than ad-hoc
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5. Discuss IPR violations in cyberspace with two relevant examples.

- **Copyright infringement / Piracy:** Unauthorized copying, distribution or streaming of copyrighted works (software, movies, music, e-books) via file-sharing sites or torrent networks. This harms creators and businesses; remedies include DMCA takedowns, civil suits, and criminal penalties in some jurisdictions.
Example: Uploading and distributing pirated software or movies on a torrent site; the site operator and uploaders can be subject to takedown requests and legal action.
 - **Trademark abuse and counterfeiting / Domain squatting:** Using identical or confusingly similar brand names, images, or domain names to sell counterfeit goods or to mislead customers (phishing/typosquatting). This damages brand reputation and misleads consumers.
Example: A fake e-commerce listing that uses a well-known brand's trademark to sell counterfeits, or registering a domain like "amaz0n-payments.com" to phish user credentials — both actionable under IPR and consumer-protection laws.
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UNIT 3

1 Mark Questions

1. Define Computer Forensics.

→ *Computer Forensics is the process of collecting, preserving, analyzing, and presenting digital evidence in a legally acceptable manner.*

2. Expand FAT and NTFS.

→ FAT – File Allocation Table; NTFS – New Technology File System.

3. Name any two UNIX file systems.

→ UFS (Unix File System), ext (Extended File System).

4. What is meant by “Data Acquisition” in forensics?

→ It is the process of collecting and imaging digital evidence from a device while ensuring its integrity.

5. Give one example of an Internet Artifact.

→ Browser history.

2 Mark Questions

1. State any two cardinal rules of computer forensics.

- **Preserve the original evidence:** Never alter the original media; always create a verified forensic image/copy for analysis.
- **Maintain chain of custody & document everything:** Record who handled the evidence, when, where and what was done to it to ensure admissibility in court.

2. Differentiate between FAT16 and FAT32.

Feature	FAT16	FAT32
Designed for / Introduced	Older FAT variant used on small volumes and early PCs.	Introduced later to support much larger disks and improve space efficiency.
Volume & file size (practical)	Suitable for smaller partitions; limited maximum volume/file sizes.	Supports much larger partitions and files (designed for modern larger disks).
Cluster size & space efficiency	Uses larger clusters on bigger volumes, causing more internal slack (wasted space).	Uses smaller clusters for large volumes, improving storage efficiency.
Root directory & structure	Fixed-size root directory (limited number of entries).	Root directory is a normal chainable cluster—no fixed-entry limit.
Compatibility / Use cases	Highly compatible with very old systems and embedded devices.	Widely used for larger removable media and older Windows systems; better for modern storage needs.

3. What is the purpose of authentication in the forensic process?

Authentication ensures that digital evidence is genuine, untampered, and legally admissible. It provides confidence that the data collected is an exact copy of the original and has not been altered during acquisition, transfer, or analysis. This is usually achieved using cryptographic hash values (like MD5 or SHA) which remain constant if the evidence is unchanged.

4. List two examples of computer artifacts.

- **Windows Registry keys** (e.g., MRU lists, user profile info).
 - **System/event logs** (e.g., Windows Event Log, syslog entries).
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5. What is the significance of OS artifacts in forensic investigation?

OS artifacts are traces left behind by the operating system, such as system logs, registry keys, temp files, prefetch data, or recent file lists. These artifacts are significant because they help investigators reconstruct user actions, verify timelines, detect unauthorized access, and identify which programs or files were recently used. They provide context that supports other evidence and are often crucial in linking suspects to activities on a device.

4 Mark Questions

1. Explain the process of data acquisition in computer forensics.

- **Authorization & preparation:** Obtain legal authority (warrant/consent), define scope and target devices, and prepare tools (write-blockers, imaging software).
- **Secure & document scene:** Photograph devices, label and log their state (powered on/off, network connections) and maintain chain of custody.
- **Collect volatile data (if live):** If the system is running, capture RAM, network connections, and running processes before shutdown.
- **Create forensic images:** Use a write-blocker to create bit-for-bit images (dd/FTK Imager/Guymager) of drives or partitions and capture copies of relevant removable media.
- **Authenticate & store:** Compute and record cryptographic hashes (MD5/SHA-1/SHA-256) of originals and images, verify hashes post-copy, and store evidence in secure, access-controlled storage with full documentation.
Example: Capture RAM with a live tool, then image the disk through a hardware write-blocker and record SHA-256 hashes for both original and image.

2. Compare FAT file system with NTFS with respect to security and efficiency.

- **Security (access control & integrity):** FAT has no built-in ACLs, encryption or journaling — minimal security; NTFS supports ACLs, file encryption (EFS), auditing and journaling (transactional metadata), so it is far more secure and forensically informative.
- **Metadata & recoverability:** FAT stores simple directory entries and a File Allocation Table (less metadata), making some carving easier but less contextual info; NTFS stores rich metadata in the MFT (timestamps, ACLs, alternate data streams), aiding attribution and timeline reconstruction.
- **Fault tolerance & recovery:** FAT (no journal) is more prone to corruption on crashes; NTFS journaling improves consistency and speeds safe recovery, though the journal can overwrite quick recovery traces.
- **Efficiency & scalability:** FAT is simple and efficient on small volumes/removable media but wastes space with large clusters on big volumes; NTFS scales to larger disks and files with better space management and performance for many small files.

Example: USB sticks often use FAT32 for cross-platform compatibility, while NTFS is preferred for Windows system disks because of security and metadata.

3. Describe the importance of Internet artifacts in forensic investigations.

- **Definition & sources:** Internet artifacts are browser and network traces (history, cache, cookies, bookmarks, download records, DNS and proxy logs, server logs, email headers) left by web activity.
- **Reconstructing activity & intent:** They provide direct evidence of visited sites, searches, downloads, and timestamps — useful to reconstruct what a user did online and whether actions were deliberate (intent).
- **Linking local to remote:** Cookies, session tokens and cached credentials can link a local machine to specific online accounts or server sessions; server logs and ISP records can tie activity to IP addresses and timeframes.
- **Evidence of exfiltration & communications:** Cache and download entries show transferred files, while network logs can reveal data exfiltration channels or communications with command-and-control servers.
Example: Browser history + server logs can show a user accessed and downloaded illicit material at a given time; cookie/session data can corroborate an authenticated session.

4. Write short notes on UNIX file systems used in forensics.

- **ext2 / ext3 / ext4 (Linux):** ext2 (no journal) allows straightforward carving; ext3 adds journaling (safer on crash but journal affects recoverability); ext4 adds extents and larger volume/file support — forensic focus is on inodes (metadata), timestamps (atime/mtime/ctime), and journal handling which affects deleted-file recovery.
 - **UFS / FFS (BSD, older UNIX):** Uses inodes and cylinder groups; file metadata lives in inodes and deletion often clears directory entries but leaves blocks until overwritten — forensic tools inspect inode tables and indirect blocks.
 - **XFS / ReiserFS:** High-performance, journaled systems used on servers (XFS), with allocation groups and advanced allocation strategies; journaling improves integrity but complicates recovery of recent deletes.
 - **Forensic implications (common):** Investigators analyze inodes, indirect blocks, and journals; they use specialized tools (e.g., The Sleuth Kit, debugfs) to extract metadata, recover deleted files, and build timelines.
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5. Explain any two forensic applications of OS artifacts with examples.

- **(A) Timeline reconstruction:** OS artifacts like file system timestamps (MFT entries on NTFS / inodes on UNIX), Windows Event Logs, macOS Unified Logs and prefetch/journal entries allow investigators to build a precise timeline of file creation, modification, execution and system events.
Example: MFT timestamps + Windows Event Log entries can show when a malicious executable was copied and later executed, establishing sequence and timeframe.
- **(B) User activity & device attribution:** Artifacts such as Windows Registry MRU lists, USBSTOR registry keys, shellbags, and mount/last-seen entries record recently used files, attached devices and folders accessed, helping to attribute actions to a user or link a removable device by serial number.
Example: USBSTOR registry entries revealing a USB drive's serial number matched to a physical drive seized from a suspect links that device to files found on the system.